UDAY KIRAN REDDY K

Data Scientist

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PROFILE SUMMARY

Data Scientist and M.S. Computer Science candidate combining 1+ years of analytics-based programming with applied AI and predictive modeling. Proficient in Python and SQL with deep knowledge of database structures. Skilled at forecasting and driving tactical solutions, demonstrated by reducing operational costs by 40% on financial platforms processing $2T+ daily. Strong written and verbal communicator adept at bridging technical and stakeholder needs.

PROFESSIONAL EXPERIENCE

Publicis Sapient, *Software Engineer* — 01/2024 – 07/2024 | Bengaluru, India

- Orchestrated automated post-trade reconciliation pipelines to address regulatory risks for $2T+ in daily transactions, ensuring strict MiFID II compliance.
- Engineered a scalable ETL framework using Python, Azure Functions, and RabbitMQ, replacing 12 manual workflows and reducing reconciliation effort by 40% (~500 annual hours saved).
- Developed real-time observability dashboards using Grafana, Azure Data Explorer (ADX), and Cosmos DB to monitor critical system health, cutting Mean Time to Detection (MTTD) by 50%.
- Spearheaded cross-functional incident response for Oracle DB outages to prevent data loss, preventing 2 critical production incidents and preserving data integrity for 25+ downstream analytics reports.

ArkaShine Innovations, *Software Developer* — 04/2023 – 10/2023 | Bengaluru, India

- Designed and deployed PyTorch-based precision irrigation models to solve resource inefficiency, utilizing IoT sensor and weather data to increase scheduling accuracy by 30% and reduce water consumption by 25%.
- Optimized the model inference pipeline via advanced hyperparameter tuning and preprocessing to enable real-time edge predictions, achieving a 3x increase in throughput and a 50% reduction in latency.
- Delivered an estimated $12,000 in annual savings for medium-sized farms by minimizing resource wastage and maximizing operational efficiency through data-driven automation.

PROJECTS

Contextual Hotel Ranking Engine (Expedia Data), *Deep Learning, TensorFlow, Learning-to-Rank*

- Engineered a Wide & Deep Neural Network with Entity Embeddings (dim=50) to resolve the "Cold Start" problem for sparse user history, achieving an NDCG@5 of 0.46 (9x baseline) by prioritizing top-rank relevance.
- Implemented a Revenue-Aware Re-Ranking layer to balance user relevance with high-margin inventory, simulating a 4% uplift in Gross Booking Value (GBV) while maintaining user satisfaction metrics.

Credit Risk & Probability Calibration Engine (Fintech), *XGBoost, SHAP, Isotonic Regression*

- Mitigated financial risk in high-stakes lending by calibrating XGBoost model scores using Isotonic Regression, reducing the Brier Score by 33% (0.21 → 0.14) to ensure "Approval Odds" were statistically reliable.
- Achieved ECOA regulatory compliance for "Black Box" AI decisions by integrating SHAP (TreeExplainer), generating automated and interpretable "Adverse Action" codes for rejected applicants.

Dynamic Pricing & Yield Management Engine (Airbnb Data), *LightGBM, Quantile Regression*

- Overcame static "average price" limitations by developing a Probabilistic Pricing Model using LightGBM with Quantile Loss, generating a 5th-95th percentile confidence interval to capture market volatility and Price Elasticity.
- Enabled data-driven Yield Management by deploying a "Smart Pricing" inference pipeline that filters for censored demand (booked-only data), allowing hosts to visually optimize Occupancy Probability vs. Profit Margin.

Clinical Text Triage & Diagnosis System (Healthcare NLP), *BERT, PyTorch, Transformers*

- Programmed the triage of unstructured patient notes to reduce "Referral Leakage" by fine-tuning a Bio-Medical BERT Transformer, achieving ~80% F1-Score on complex, multi-symptom medical cases.
- Tackled complex medical lexicon (e.g., "Patella Fracture") using Transfer Learning and WordPiece tokenization, deploying a real-time inference API that serves predictions with confidence scores for human-in-the-loop validation.

PUBLICATIONS

Machine Learning Powered Car Recommendation System, *Springer Nature (SmartCom 2024)*

- Authored a hybrid recommender engine combining Random Forest and Collaborative Filtering, which outperformed baseline Neural Network architectures by 18% in precision metrics.
- Published in Lecture Notes in Networks and Systems (DOI: 10.1007/978-981-97-1326-4_28).

EDUCATION

M.S. in Computer Science (Data Science), *University of North Carolina at Charlotte* — 2025 – 2026 | GPA: 3.77/4.00

B.Tech in Computer Science, *M. S. Ramaiah University of Applied Sciences, India* — 2020 – 2024 | GPA: 8.91/10.00

AREAS OF EXPERTISE

Python, **SQL** (PostgreSQL, BigQuery), **Bash**, **Git**, **XGBoost**, **LightGBM**, **Random Forest**, **Gradient Boosting**, **Linear/Logistic Regression**, **K-means Clustering**, **PCA**, **Pytorch**, **TensorFlow/Keras**, **Transfomers**, **LLMs**, **Transfer Learning**, **Time-Series Forecasting**, **NLP**, **Recommender Systems**, **A/B Testing**, **ETL/ELT Pipeline**, **Docker**, **Kubernetes**, **CI/CD**, **Azure**, **AWS**, **Airflow**, **RabbitMQ**, **Pandas**, **NumPy**, **Scikit-Learn**, **Tableau**, **Grafana**, **Matplotlib**, **Seaborn**, **Regulatory Compliance**, **Hadoop**, **Spark**, **Snowflake**, **DAGS**, **DBT**, **Uplift Modeling**, **SAS**